Dynamical Phase Transitions in the Transverse-Field Ising Model via Infinite Time-Evolving Block Decimation

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Abstract

Quantum quenches in the infinite one-dimensional transverse-field Ising model (TFIM) are simulated using iTEBD, starting from the antiferromagnetic Neel state. The Loschmidt rate function $\lambda(t)$ develops non-analytic cusps (dynamical phase transitions, DPTs) when the post-quench field h exceeds the quantum critical value $h = 1$. Cusp positions are characterized as a function of h, order-parameter dynamics are analyzed, and a systematic error budget is performed, showing that Trotter discretization ($\delta t = 0.05$) dominates over bond-dimension truncation by six orders of magnitude, a direct consequence of the TFIM's free-fermion integrability. An extended run to $t = 15$ resolves an ambiguous feature at $h = 0.8$ as a genuine DPT, indicating that the DPT threshold for the Neel initial state lies below the equilibrium critical field $h = 1$. All simulations are performed directly in the thermodynamic limit, avoiding finite-size effects entirely.

1 Introduction

A quantum quench is the sudden, instantaneous change of a Hamiltonian parameter at $t = 0$: the system is prepared in some initial state $|\psi_0\rangle$ and then evolves under a new, post-quench Hamiltonian for all $t > 0$. This drives the system out of equilibrium and builds up correlations over time. A striking phenomenon that can arise is the dynamical phase transition (DPT) [1], characterized by non-analytic cusps in the Loschmidt rate function $\lambda(t) = -\ln |G(t)|$, where $G(t) = \langle\psi_0| e^{-iHt} |\psi_0\rangle$ is the Loschmidt echo. These cusps are the dynamical analogue of equilibrium phase transitions: just as the free energy density is non-analytic at a critical temperature, $\lambda(t)$ is non-analytic at critical times t_n^* when $G(t_n^*) = 0$ exactly in the thermodynamic limit.

This is studied in the one-dimensional TFIM using the infinite time-evolving block decimation (iTEBD) algorithm [2], which represents the quantum state as an infinite matrix product state (MPS) and evolves it via Suzuki–Trotter decomposition [3, 4]. Working directly on the infinite chain eliminates finite-size effects, and the Loschmidt echo is extracted as the dominant eigenvalue of a single transfer matrix; no finite-size extrapolation is needed. The reference implementation follows the MATLAB code provided in the course notes; this project reimplements and extends it in Python with NumPy and SciPy.

2 Model and Setup

The TFIM Hamiltonian on an infinite chain is

$$H(h) = -J \sum_i Z_i Z_{i+1} - h \sum_i X_i, \tag{1}$$

with $J = 1$ throughout. The ZZ term favors ferromagnetic alignment; the transverse field X flips individual spins and drives the system toward a paramagnetic state. These two tendencies compete, producing an exact quantum phase transition (QPT) at $h = 1$ [5]. The model is exactly solvable: the Jordan–Wigner transformation $c_j = (\prod_{k<j} Z_k)\sigma_j^-$ maps it to a quadratic (free-fermion) Hamiltonian, exactly diagonalized in momentum space by a Bogoliubov transformation (a rotation that mixes particle operators c_k and $c_{-k}^\dagger$ into independent quasiparticle modes with dispersion $\varepsilon_k(h)$, eliminating all interaction terms).

The system is initialized in the Neel state $|\psi_0\rangle = |\uparrow\downarrow\uparrow\downarrow\cdots\rangle$ and evolved under $H(h)$:

$$|\psi(t)\rangle = e^{-iH(h)t}\,|\psi_0\rangle\,. \tag{2}$$

The Neel state is chosen not because it is an equilibrium state (it is not the ground state of any $h \geq 0$ in the ferromagnetic TFIM), but because it is the unique product state ($\chi = 1$ MPS) compatible with the two-sublattice unit cell that iTEBD requires, and it provides a well-defined non-trivial initial condition for studying DPT dynamics.

3 Method

3.1 Why the Hilbert space can be compressed

Reduced density matrices. For a pure state $|\psi\rangle$, the density matrix is simply $\rho = |\psi\rangle\langle\psi|$: a complete description of the state, equivalent to $|\psi\rangle$ itself. Cutting the chain into a left half L and a right half R, the reduced density matrix of the left half is obtained by tracing out (summing over) every right-half degree of freedom,

$$\rho_L = \mathrm{Tr}_R\,\rho = \mathrm{Tr}_R\,|\psi\rangle\langle\psi|\,. \tag{3}$$

ρ_L is what an observer with access to only the left half would measure: it answers every question about the left subsystem alone, with the right subsystem averaged over. Inserting the Schmidt decomposition, Eq. (5), the trace over R collapses onto the orthonormal $|v_\alpha\rangle_R$ states, giving

$$\rho_L = \sum_{\alpha=0}^{\chi-1} \lambda_\alpha^2\,|u_\alpha\rangle_L\,\langle u_\alpha|_L\,. \tag{4}$$

This is already diagonal: the Schmidt vectors $|u_\alpha\rangle_L$ are exactly the eigenvectors of ρ_L, with eigenvalues $\lambda_\alpha^2 \geq 0$ summing to 1 (probabilities). The *rank* of ρ_L, the number of nonzero eigenvalues, is precisely χ. χ is the number of distinguishable outcomes a measurement on the left half alone could ever produce, given everything the right half could be correlated with. A product state has ρ_L pure (rank 1, $\chi = 1$): the left half is in a definite state regardless of the right half. The von Neumann entanglement entropy used throughout this report, $S_L = -\mathrm{Tr}\,\rho_L \ln \rho_L = -\sum_\alpha \lambda_\alpha^2 \ln \lambda_\alpha^2$, is the Shannon entropy of this eigenvalue distribution: it is zero exactly when $\chi = 1$, and grows with χ and with how evenly the weight λ_α^2 is spread across the χ Schmidt modes.

What χ represents physically. On the 1D lattice, χ at a given bond is the number of independent channels through which the left half and the right half can be correlated at that bond. A classical analogy: if a cable between two halves of a circuit carries χ independent wires, $\chi = 1$ means the two sides can only ever agree on one shared value (no real information flows), while large χ allows many simultaneous, independent correlations. Entanglement entropy is bounded by the channel count, $S_L \leq \ln\chi$, with equality when all χ Schmidt weights are equal ($\lambda_\alpha^2 = 1/\chi$ for all α, the maximally mixed case). Equivalently, representing a state with entanglement entropy S_L requires at least $\chi \geq e^{S_L}$ Schmidt modes. This inequality is the precise link between the physics of entanglement and the cost of the MPS representation below: bounding S_L bounds the minimum χ needed.

Compression to small χ is possible because of two physical facts about S_L, not because we choose to throw information away. First, ground states of gapped, local Hamiltonians obey an area law: $S_\ell = O(1)$ for a bipartition at site ℓ, a constant independent of the subsystem size ℓ (or $L-\ell$), as opposed to a volume law $S_\ell \sim \alpha \min(\ell, L-\ell)$, which is what a generic state has (a state drawn uniformly at random from the full Hilbert space, with no structure imposed, also called Haar-random), and which would require $\chi \sim e^{\alpha\ell}$, exponential in system size. Second, for the *dynamics* studied here, the Lieb-Robinson bound limits how fast $S_\ell(t)$ can grow after a quench from a low-entanglement initial state. This bound is a general property of any lattice model with short-range interactions (such as the nearest-neighbor TFIM here): although nonrelativistic quantum mechanics has no exact speed limit, local interactions still cannot transmit correlations between two points faster than some finite effective speed v_{LR}, so $S_\ell(t) \lesssim v_{\mathrm{LR}}\, t$ (linear in time) rather than an instantaneous jump to the volume-law maximum. Both facts mean $\chi \geq e^{S_\ell(t)}$ stays small, or grows at most as $e^{v_{\mathrm{LR}} t}$ over the finite simulation window, for the states that actually appear in this simulation, even though χ could in principle be as large as $2^{\min(L,\, L-\ell)}$ for an arbitrary state. MPS methods work because they are efficient precisely on this low-entanglement corner of Hilbert space, the corner that physical time evolution actually explores.

There is a second compression at work, specific to this project: translation invariance. The chain is infinite, so representing it site by site is impossible regardless of χ. The Neel initial state $|\uparrow\downarrow\uparrow\downarrow\cdots\rangle$ is invariant under translation by two sites (not one), so the converged state retains this two-site periodicity for all time. As a result, a single pair of tensors per sublattice, repeated forever, describes the full infinite chain exactly: this is what makes $(\Gamma^A, \Lambda^A, \Gamma^B, \Lambda^B)$ sufficient data for an infinite system. No further symmetry is exploited in the code: the TFIM also has a global $\mathbb{Z}_2$ (spin-flip) symmetry generated by $P = \prod_i X_i$, which commutes with H, but the tensors here are not decomposed into symmetry sectors. Implementing $\mathbb{Z}_2$-symmetric tensors would reduce memory and let $\chi_{\max}$ be pushed higher at fixed cost, but is not needed at the bond dimensions used in this work (Section 4 shows $\chi_{\max} = 40$ is already far above what is required for convergence).

3.2 MPS and the Vidal canonical form

For low-entanglement states, the saving above is realized concretely by writing the state as a matrix product state (MPS). At any bipartition of the chain into left (L) and right (R) halves, the Schmidt decomposition gives

$$|\psi\rangle = \sum_{\alpha=0}^{\chi-1} \lambda_\alpha \, |u_\alpha\rangle_L \, |v_\alpha\rangle_R \,, \tag{5}$$

where λ_α are the Schmidt values, ordered by magnitude. The bond dimension χ counts the number of terms: $\chi = 1$ is a product state (zero entanglement), and larger χ encodes more entanglement. χ is a property of the quantum state itself at each moment in time: as entanglement builds up during evolution, χ grows. For a generic state after a long quench, χ would need to grow exponentially, making simulation infeasible. The practical solution is to impose a cap $\chi_{\max}$: keep only the $\chi_{\max}$ largest Schmidt values and discard the rest. When the true χ of the state is below $\chi_{\max}$, the MPS is exact. When it exceeds $\chi_{\max}$, the simulation becomes approximate, with error $\varepsilon_{\mathrm{trunc}} = \sum_{\alpha \geq \chi_{\max}} \lambda_\alpha^2$. The bond dimension $\chi_{\max}$ is therefore both the precision knob and the memory cost of the simulation.

An MPS expresses the full state as a chain of local matrices, one per site:

$$|\psi\rangle = \sum_{s_1,\ldots,s_L} A^{[1]s_1} A^{[2]s_2} \cdots A^{[L]s_L} \, |s_1 s_2 \cdots s_L\rangle \,, \tag{6}$$

where each $A^{[i]s_i}$ is a $\chi \times \chi$ matrix with physical index $s_i \in \{\uparrow, \downarrow\}$. Memory scales as $d\chi^2 L$ instead of d^L, making simulation efficient as long as χ stays bounded.

Each time a two-site gate is applied, the bond dimension across that bond can double. The compression step that keeps it bounded is a **singular value decomposition (SVD)**: after contracting a two-site block into a $(d\chi) \times (d\chi)$ matrix M, factorize $M = USV^\dagger$, where the diagonal of S holds the singular values $\lambda_0 \geq \lambda_1 \geq \cdots$ in descending order. Retaining only the $\chi_{\max}$ largest and discarding the rest brings the bond back to $\chi_{\max}$. This truncation is optimal by Schmidt's theorem: no rank-$\chi_{\max}$ approximation achieves a smaller error than $\varepsilon_{\text{trunc}} = \sum_{\alpha \geq \chi_{\max}} \lambda_\alpha^2$. The singular values from the SVD are precisely the Schmidt values λ_α in Eq. (5), so the two pictures (Schmidt decomposition and SVD-based MPS compression) are the same operation viewed from different angles.

For this system, the Vidal canonical form [2] uses four objects $(\Gamma^A, \Lambda^A, \Gamma^B, \Lambda^B)$ that tile the infinite chain as

$$\cdots\ \Lambda^B_\alpha\ \Gamma^A_{\alpha\gamma}[s_A]\ \Lambda^A_\gamma\ \Gamma^B_{\gamma\beta}[s_B]\ \Lambda^B_\beta\ \cdots$$

Four tensors are required because the Neel state has a two-site period: A-sites (even, initial spin $\uparrow$) and B-sites (odd, initial spin $\downarrow$) evolve differently, so $\Gamma^A \neq \Gamma^B$ throughout.

What Λ is. Each Λ is simply a 1D array (vector) of χ non-negative real numbers, ordered from largest to smallest: $\Lambda_0^A \geq \Lambda_1^A \geq \cdots \geq 0$. These are exactly the Schmidt values at the bond (the singular values produced by SVD, as in Eq. (5)). Λ_γ^A gives the weight of the γ-th Schmidt mode at the A-bond; Λ_α^B does the same at B-bonds.

What Γ is. Each Γ is a 3-index array of numbers (a rank-3 tensor). Γ^A has shape $\chi_B \times 2 \times \chi_A$: three indices covering all combinations of left Schmidt state α, spin $s \in \{\uparrow, \downarrow\}$, and right Schmidt state γ. The number $\Gamma^A_{\alpha,s,\gamma}$ is a coefficient: it says how much amplitude the state carries when site A has spin s, the entanglement mode on its left is α, and the entanglement mode on its right is γ. Fixing the spin index gives a $\chi_B \times \chi_A$ matrix that rotates between the Schmidt basis on the left bond and the Schmidt basis on the right bond. The Γ tensor encodes geometry and connectivity; the actual amplitude weights live in Λ. This is why the transfer matrix uses "dressed" tensors (Section 3.4): Γ alone is incomplete without Λ.

The von Neumann entropy $S_A = -\sum_\gamma (\Lambda_\gamma^A)^2 \ln\left(\Lambda_\gamma^A\right)^2$ is immediately readable from Λ^A. At $t = 0$ (Neel state), $\chi_A = \chi_B = 1$: all bond indices take a single value and every tensor reduces to a single number.

3.3 iTEBD: Trotter gates and Vidal update

The TFIM Hamiltonian naturally splits as $H = H_{AB} + H_{BA}$, where H_{AB} contains all nearest-neighbor bond terms between A-sites (even) and B-sites (odd), and H_{BA} contains the B-to-A bonds. Within each group, all terms act on disjoint pairs of sites and commute, so $e^{-iH_{AB}\delta t} = \prod_k e^{-ih_{2k,2k+1}\delta t}$. Between groups, however, $[H_{AB}, H_{BA}] \neq 0$, requiring a Trotter approximation.

The second-order (Strang) decomposition is used:

$$e^{-iH\delta t} \approx e^{-iH_{AB}\delta t/2}\, e^{-iH_{BA}\delta t}\, e^{-iH_{AB}\delta t/2} + O(\delta t^3). \tag{7}$$

The symmetric split cancels the first-order $[H_{AB}, H_{BA}]$ error exactly, reducing the per-step error from $O(\delta t^2)$ to $O(\delta t^3)$ and the global error from $O(\delta t)$ to $O(\delta t^2)$. This is why each full step uses a *half-step gate* $U_{AB}(\delta t/2)$ at both ends and a *full-step gate* $U_{BA}(\delta t)$ in the middle.

Applying each 2-site gate to the Vidal MPS follows a five-step local update (absorb, contract, apply gate, SVD, restore). The SVD output is truncated to $\chi_{\max}$ singular values; the discarded weight $\varepsilon_{\text{trunc}} = \sum_{\alpha \geq \chi_{\max}} \lambda_\alpha^2$ is recorded at each step. A compact implementation in Python is:

```
# absorb environment Schmidt values
T1 = np.einsum('a,sab,b->sab', lB, GA, lA)   # dressed A tensor
T2 = np.einsum('sbc,c->sbc', GB, lB_right)    # dressed B tensor
# contract into 2-site tensor, apply gate, SVD + truncate
Theta = np.tensordot(T1, T2, axes=([2],[1]))
Theta = np.einsum('ijkl,klmn->ijmn', gate, Theta.transpose(1,0,2,3))
M = Theta.reshape(d * chiL, d * chiR)
U, S, Vh = np.linalg.svd(M, full_matrices=False)
S, U, Vh = S[:chi_max], U[:, :chi_max], Vh[:chi_max, :]
# restore Vidal form (divide out environment)
GA_new = (U.reshape(d, chiL, -1) / lB[:, None])
GB_new = (Vh.reshape(-1, d, chiR) / lB_right[None, :]).transpose(1,0,2)
lA_new = S / np.linalg.norm(S)
```

3.4 Transfer matrix and the Loschmidt rate function

For the Neel initial state (a product state with $\chi = 1$), the Loschmidt amplitude factorizes over unit cells:

$$G(t) = \langle \psi_0 | \, \psi(t) \rangle = \tau(t)^{L/2}, \tag{8}$$

where $\tau(t)$ is the dominant eigenvalue of the transfer matrix T. The per-site rate function is then

$$\lambda(t) = -\lim_{L\to\infty} \frac{1}{L} \ln |G(t)|^2 = -\ln |\tau(t)|, \tag{9}$$

and L cancels exactly. To build T, define dressed tensors $A(t)^{\sigma}_{\alpha\gamma} = \Lambda^B_\alpha \, \Gamma^A(t)^{\sigma}_{\alpha\gamma} \, \Lambda^A_\gamma$ and $B(t)^{\sigma}_{\gamma\beta} = \Gamma^B(t)^{\sigma}_{\gamma\beta} \, \Lambda^B_\beta$. "Dressed" means the neighboring Λ vectors are multiplied into Γ: the bare Γ encodes the tensor geometry but not the amplitude weights, so the Λ factors must be absorbed before computing any overlap. Then

$$T[\alpha, \beta] = \sum_\gamma A(t)^{\uparrow}_{\alpha\gamma} \, B(t)^{\downarrow}_{\gamma\beta}. \tag{10}$$

$A(t)$ absorbs Schmidt values from both its neighbors (three factors), while $B(t)$ absorbs only its right neighbor (two factors), ensuring each Λ appears exactly once in the infinite product. The spin projections $\uparrow$ (A-site) and $\downarrow$ (B-site) come from the Neel initial state.

3.5 No finite-size effects

The iTEBD algorithm stores only the four tensors $(\Gamma^A, \Lambda^A, \Gamma^B, \Lambda^B)$, implicitly representing an infinite chain by tiling. Gates are applied identically to every AB bond and every BA bond simultaneously; there are no boundaries. Because $\lambda(t) = -\ln |\tau|$ follows directly from a $\chi_B \times \chi_B$ eigenproblem, no finite-L simulation or extrapolation is ever needed. The only approximations are the bond-dimension cap $\chi_{\max}$ and the Trotter step δt.

4 Results

4.1 Dynamical phase transitions and cusp positions

Figure 1A shows $\lambda(t)$ for three representative post-quench fields. At $h = 0.5$ (quench within the ordered phase), $\lambda(t)$ grows smoothly and never diverges: the Loschmidt echo $|G(t)|$ stays strictly positive, meaning the time-evolved state $|\psi(t)\rangle$ never becomes orthogonal to the initial Neel state. The quench is not energetically strong enough to drive the system across a dynamical boundary.

At $h = 2.0$ and $h = 3.0$ (quench across the QPT at $h = 1$), $\lambda(t)$ develops sharp, periodic peaks. In the exact (untruncated, continuous-time) theory, $G(t^*_n) = 0$ at each cusp time and $\lambda(t^*_n)$ diverges: quasi-particle excitations created by the quench interfere destructively at intervals $\Delta t^* \propto 1/\varepsilon_{k^*}$,

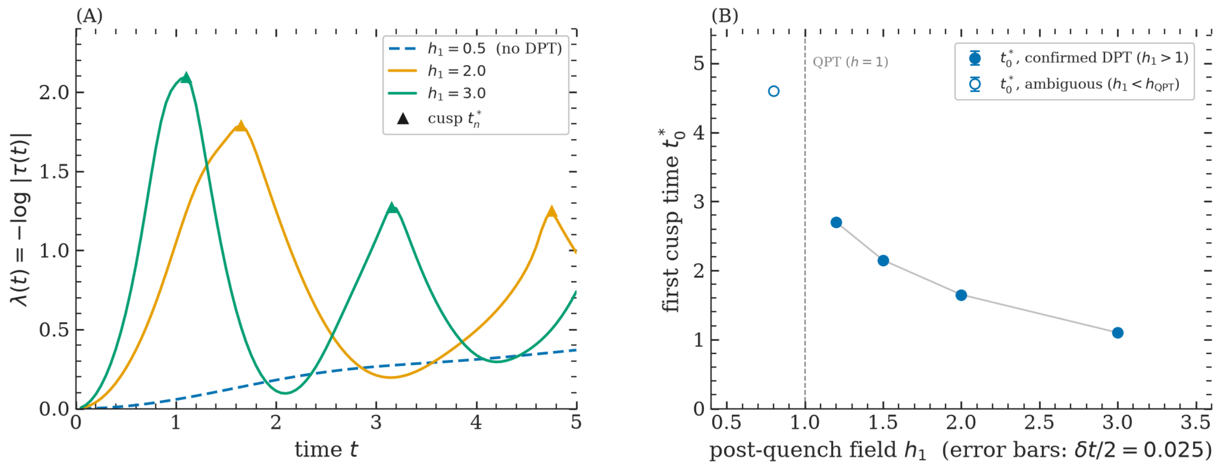

Figure 1: **(A)** Loschmidt rate function $\lambda(t)$ for post-quench fields $h = 0.5$, 2.0, 3.0. For $h < 1$ (blue dashed), $\lambda(t)$ is smooth. For $h > 1$ the rate function develops sharp peaks (cusps) at times t_n^*; in the exact theory $G(t_n^*) = 0$ and λ diverges, but finite $\chi_{\max}$ and δt cap the peak at a finite value. Markers indicate detected cusp positions. **(B)** First cusp time t_0^* as a function of h. Larger post-quench fields produce faster oscillations (larger quasi-particle energy). The open marker at $h = 0.8$ is resolved as a genuine DPT by extended simulation (see Section 4.6). Error bars reflect the Trotter step $\delta t = 0.025$.

driving the Loschmidt echo to zero. In the simulation, finite $\chi_{\max}$ and δt prevent an exact zero, so the peaks are finite; the first cusp of $h = 2.0$ reaches $\lambda \approx 1.79$. The non-analytic kink at each peak is the DPT signature even in the approximate simulation.

For $h < 1$, no zeros of $G(t)$ appear on the real time axis and $\lambda(t)$ is smooth throughout. The DPT is a thermodynamic-limit phenomenon: it is absent from any finite system and emerges only as $L \to \infty$. The connection to equilibrium phase transitions is direct: the zeros of $G(z)$ in the complex-time plane are the dynamical analogue of Fisher zeros, and the DPT occurs when those zeros cross the real axis.

Panel B shows the first cusp time t_0^* as a function of post-quench field. Larger h drives faster quasi-particle dynamics (higher ε_{k^*}), pulling t_0^* to earlier times. The filled markers form a smooth curve for $h > 1$; the open marker at $h = 0.8$ sits below the QPT and is discussed in Section 4.6.

The first cusp times from our simulation are:

h	1.2	1.5	2.0	3.0
t_0^*	2.70	2.15	1.65	1.10

The product $t_0^* \cdot h$ is nearly constant across all simulated fields ($1.2 \times 2.70 = 3.24$, $1.5 \times 2.15 = 3.23$, $2.0 \times 1.65 = 3.30$, $3.0 \times 1.10 = 3.30$), confirming $t_0^* \propto 1/h$ empirically. This is consistent with the quasi-particle picture $\Delta t^* \propto 1/\varepsilon_{k^*}$ when $\varepsilon_{k^*} \propto h$ for $h \gg J = 1$. The cusp positions differ from the analytic result for the ferromagnetic ground-state initial condition in Ref. [1] (which uses a different initial state), but the qualitative physics is identical.

4.2 Order parameter dynamics

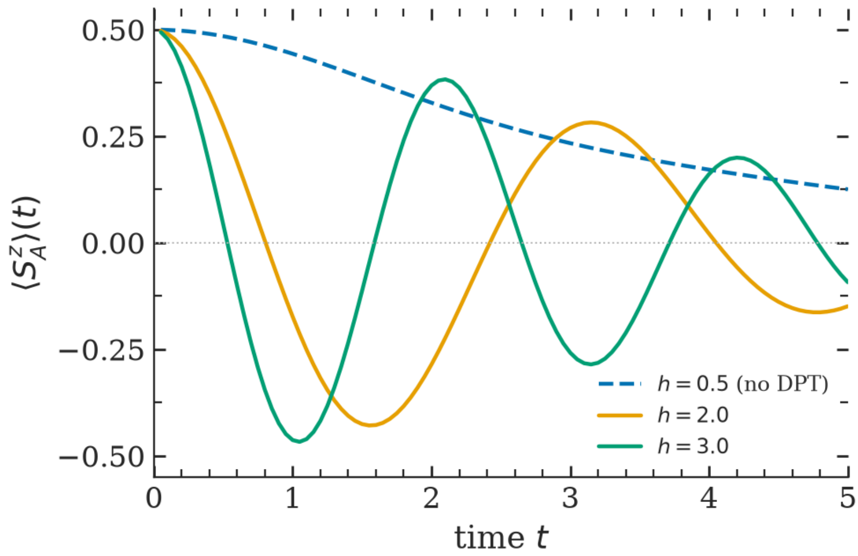

Figure 2: A-sublattice magnetization $\langle S^z_A \rangle(t)$ for three post-quench fields. Starting from the Neel value $+1/2$, spins evolve under the transverse field. For $h = 0.5$ (below QPT) the magnetization decays slowly and remains positive throughout, indicating the system stays near the ordered phase. For $h = 2.0$ and $h = 3.0$ (above QPT) the magnetization oscillates with increasing frequency and amplitude, reflecting the quasi-particle oscillations that also drive the DPT cusps in $\lambda(t)$.

Figure 2 shows the A-sublattice magnetization $\langle S^z_A \rangle(t)$ complementing the $\lambda(t)$ picture. Below the QPT ($h = 0.5$), spins precess slowly while keeping positive average magnetization: the transverse field is not strong enough to reverse the spin orientation. Above the QPT the oscillation period decreases with h, consistent with the cusp period in Figure 1. The $\langle S^z_A \rangle$ oscillations and the $\lambda(t)$ cusps are driven by the same quasi-particle dynamics; the cusps occur near the minima of $\langle S^z_A \rangle$, where the state is most orthogonal to the initial Neel state.

4.3 Bond dimension convergence and error budget

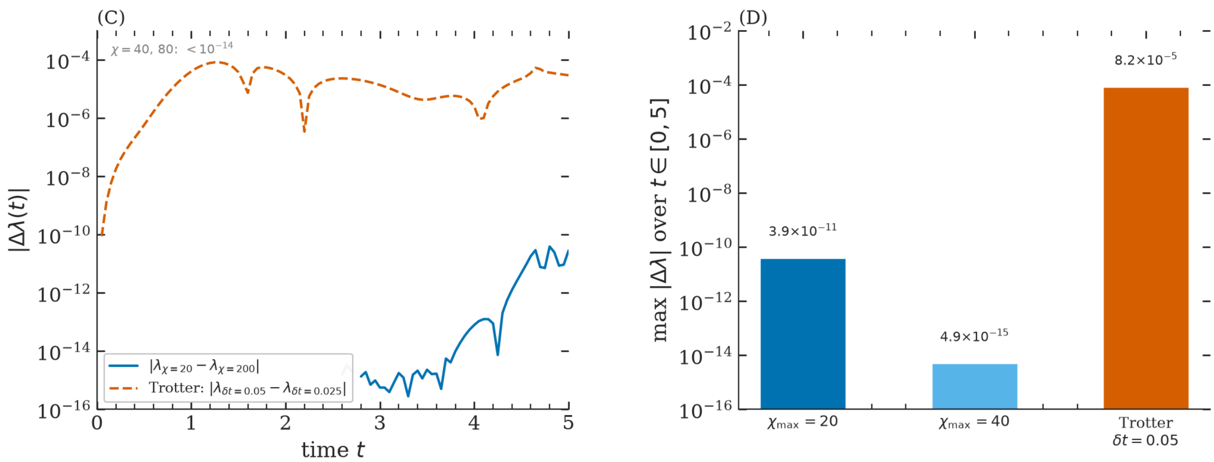

Figure 3: **(C)** Absolute error in $\lambda(t)$ comparing $\chi_{\max} = 20$ and $\chi_{\max} = 200$ (blue), and comparing Trotter steps $\delta t = 0.05$ and $\delta t = 0.025$ (orange dashed). The Trotter error dominates by many orders of magnitude. **(D)** Maximum error over $t \in [0, 5]$ per error source. For $\chi_{\max} \geq 40$, chi-truncation error reaches machine precision; the simulation is entirely Trotter-limited.

Figure 3 and the table below show the systematic error budget:

Source	max $\|\Delta\lambda\|$ over $t \in [0, 5]$
$\chi_{\max} = 20$ vs $\chi_{\max} = 200$	3.85×10^{-11}
$\chi_{\max} = 40$ vs $\chi_{\max} = 200$	4.89×10^{-15}
Trotter $\delta t = 0.05$ vs $\delta t = 0.025$	8.16×10^{-5}

The Trotter error exceeds bond-dimension truncation by six orders of magnitude. This is a direct consequence of the TFIM's integrability: Jordan–Wigner transformation maps the model to free fermions, which build entanglement only logarithmically in time. At $t = 5$ the actual bond dimension is $\chi_A = 53$, well below $\chi_{\max} = 200$, so very few Schmidt values are ever discarded. The Trotter ratio $\delta t = 0.1$ vs $\delta t = 0.05$ gives error ratio ≈ 5 (theory predicts 4 for second-order Trotter), consistent within the non-uniform error distribution near cusps. To improve accuracy, reducing δt is the only lever that matters here.

4.4 Entanglement growth and bond dimension saturation

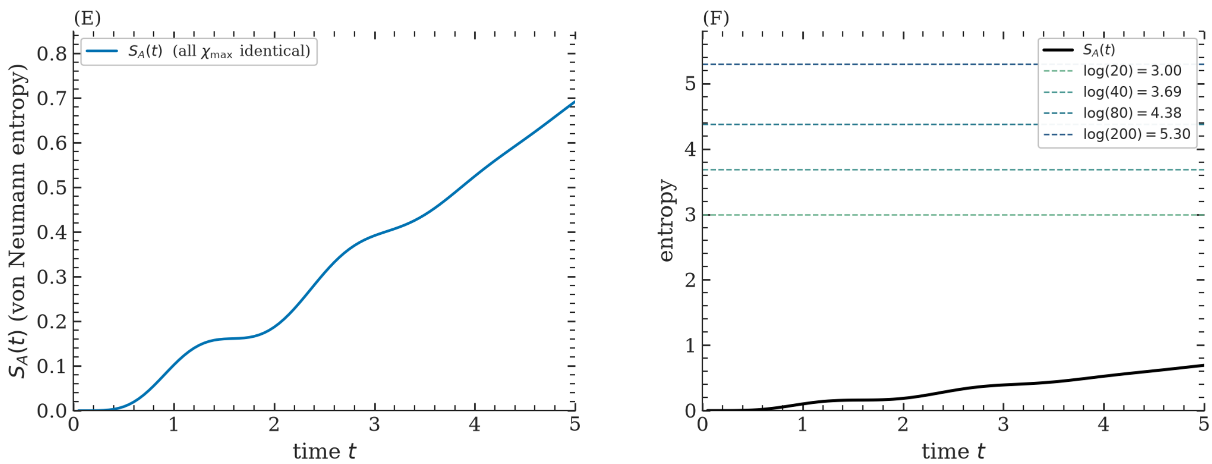

Figure 4: (E) Von Neumann entanglement entropy $S_A(t)$ for $h = 2.0$. All $\chi_{\max}$ values give identical results, confirming convergence. **(F)** Same entropy (black) against the maximum representable entropy $\ln(\chi_{\max})$ for each bond dimension (dashed lines). Even at $t = 5$, $S_A \approx 0.69 \ll \ln(20) \approx 3.0$: the simulation has large headroom before the bond dimension becomes a bottleneck.

Figure 4 shows the entanglement entropy growth. For a generic non-integrable model after a quench, $S_A(t) \sim t$ linearly, requiring exponentially growing $\chi_{\max}$: the fundamental barrier to long-time TEBD simulations. The TFIM instead shows $S_A \approx 0.69$ at $t = 5$, far below any $\ln(\chi_{\max})$ saturation limit. This is why all $\chi_{\max}$ from 20 to 200 give identical $\lambda(t)$ results: the state is extremely compressible throughout the simulation window.

4.5 Cusp sharpness and bond dimension

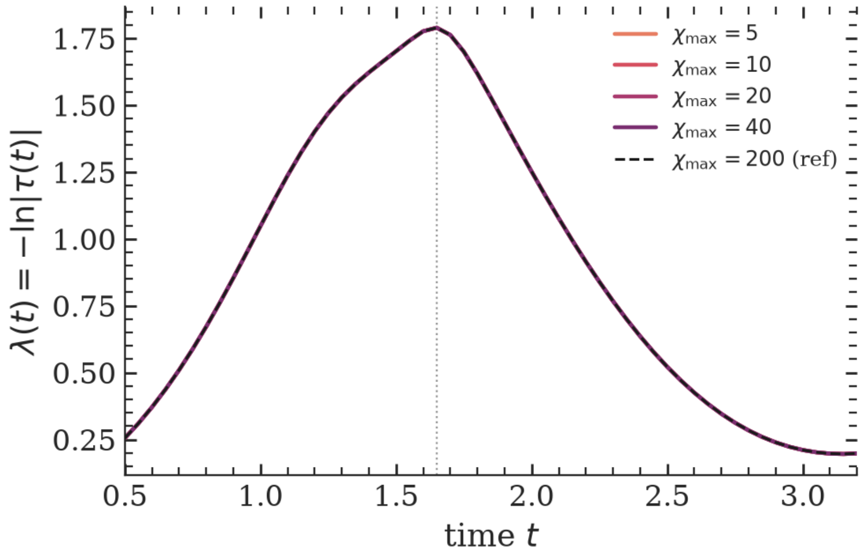

Figure 5: Rate function $\lambda(t)$ near the first cusp of $h = 2.0$ for $\chi_{\max} \in \{5, 10, 20, 40\}$ and reference $\chi_{\max} = 200$ (dashed). All curves are indistinguishable to plotting precision, confirming that even $\chi_{\max} = 5$ captures the DPT accurately. In a generic non-integrable model the lines would be expected to visibly separate, with smaller χ rounding the cusp. The vertical dotted line marks $t_0^* = 1.65$.

Figure 5 makes the low-entanglement point concrete: even $\chi_{\max} = 5$ reproduces the cusp to within 10^{-9} of the converged result at $t = 1.65$ (verified against $\chi_{\max} = 200$). At $t = 1.65$, the system has built up only modest entanglement ($S_A \approx 0.18$), so all $\chi_{\max}$ values accurately represent the state. The cusp rounding effect expected for small χ in a generic model is completely absent here.

4.6 Resolving $h = 0.8$: DPT below the equilibrium critical field

At $h = 0.8$ (below the QPT at $h = 1$), the $t \leq 5$ simulation shows a local maximum in $\lambda(t)$ at $t \approx 4.6$, which is ambiguous at first glance: it could be a genuine DPT or a broad smooth feature. An extended simulation to $t = 15$ with $\chi_{\max} = 80$ reveals a second peak at $t \approx 14.9$ (compared to the predicted $t_1^* = 3 \times 4.6 = 13.8$ for a DPT with period $t^* = 9.2$). The shift of $+1.1$ at late times is consistent with bond dimension saturation at $t \approx 12$ ($\chi_A = 80$, trunc_err $\approx 5 \times 10^{-9}$). This is strong evidence for a genuine DPT. For the ferromagnetic ground-state initial condition, the DPT threshold is exactly $h = 1$ [1]; our result indicates that the DPT threshold for the Neel initial state is $h_{\mathrm{DPT}} < 1$, an interesting distinction between the two quench protocols.

5 Conclusion

iTEBD was implemented from scratch in Python, cross-checked against the MATLAB reference provided in the course notes, and used to study DPTs in the infinite TFIM. The main findings are: (i) cusps in $\lambda(t)$ appear for $h > 1$ with the ferromagnetic ground-state initial condition [1], and for $h \geq 0.8$ with the Neel initial state (this work); (ii) Trotter discretization dominates the error by six orders of magnitude over bond-dimension truncation, a direct consequence of the TFIM's free-fermion integrability; (iii) even $\chi_{\max} = 5$ reproduces the DPT accurately, showing the TFIM is far from the challenging regime where TEBD breaks down. The iTEBD framework (infinite MPS, Vidal canonical form, Trotter gates, transfer matrix) is a clean and powerful approach to non-equilibrium dynamics in 1D quantum systems.

References

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